

Preconditioned SGD on a Stochastic Quadratic

1 The stochastic quadratic

Fix $x_0 \in \mathbb{R}^n$ and consider

$$f(x; g, H) = g'(x - x_0) + \frac{1}{2} (x - x_0)' H (x - x_0),$$

where the pair (g, H) is random, with mean gradient $\bar{g} = \mathbb{E}[g]$ and mean Hessian $\bar{H} = \mathbb{E}[H] \succ 0$.

Because expectation is linear, the averaged objective $\mathbb{E}f$ is itself a quadratic with Hessian $\bar{H}$, and its gradient is

$$\nabla \mathbb{E}f(x) = \bar{g} + \bar{H}(x - x_0).$$

It is minimized at

$$x^\star = x_0 - \bar{H}^{-1} \bar{g}.$$

The stochastic gradient of f is

$$\nabla f(x; g, H) = g + H(x - x_0).$$

To find $x^\star$, we'll consider a preconditioner $P \succ 0$ and a step size $\eta > 0$. The preconditioned stochastic gradient update under these conditions is

$$x_{k+1} = x_k - \eta P \nabla f(x_k; g_k, H_k) = x_k - \eta P [g_k + H_k(x_k - x_0)].$$

2 The error recurrence

Define the error $e_k \equiv x_k - x^\star$ and the offset $d \equiv x^\star - x_0$. Let $\Sigma \equiv \text{Cov}(g)$ be the gradient covariance and $\tilde{H}_k \equiv H_k - \bar{H}$ the Hessian noise. Write

$$A \equiv I - \eta P \bar{H}, \quad N \equiv \mathbb{E}[\tilde{H}_k P^2 \tilde{H}_k] \succeq 0, \quad \rho \equiv \|A\|_2^2 + 2\eta^2 \|N\|_2.$$

Theorem 1. *If $\rho < 1$, then for every $k \geq 0$,*

$$\mathbb{E}\|e_k\|^2 \leq \rho^k \mathbb{E}\|e_0\|^2 + \frac{1 - \rho^k}{1 - \rho} \eta^2 (\text{tr}(P \Sigma P) + 2 d' N d).$$

So $\mathbb{E}\|e_k\|^2$ decays geometrically at rate ρ to a floor at most $\eta^2 (\text{tr}(P \Sigma P) + 2 d' N d) / (1 - \rho)$.

The floor is fluctuation rather than a persistent offset. That's because the mean error vanishes, $\mathbb{E}e_k \rightarrow 0$. Taking expectations of (1) gives $\mathbb{E}e_{k+1} = A \mathbb{E}e_k$. Hence $\|\mathbb{E}e_k\| \leq \|A\|_2^k \|\mathbb{E}e_0\| \rightarrow 0$ when $\|A\|_2 < 1$.

Proof. Write the gradient noise $\tilde{g}_k \equiv g_k - \bar{g}$, with covariance Σ ; it and $\tilde{H}_k$ are zero mean and independent of each other and of e_k . In error coordinates $x_k - x_0 = e_k + d$, and the averaged gradient is $\nabla \mathbb{E}f(x_k) = \bar{g} + \bar{H}(x_k - x_0) = \bar{H}e_k$, because $\bar{H}d = -\bar{g}$ (as $d = x^* - x_0 = -\bar{H}^{-1}\bar{g}$). The stochastic gradient is that mean plus the two noise terms,

$$\nabla f(x_k; g_k, H_k) = \nabla \mathbb{E}f(x_k) + \tilde{g}_k + \tilde{H}_k(x_k - x_0) = \bar{H}e_k + \tilde{g}_k + \tilde{H}_k(e_k + d).$$

Substituting into the update gives the recurrence

$$e_{k+1} = A e_k - \eta P \tilde{g}_k - \eta P \tilde{H}_k(e_k + d). \quad (1)$$

Condition (1) on e_k and expand $\|e_{k+1}\|^2$. Every cross term drops, because both noises are zero mean given e_k and independent of each other, leaving the sum of the deterministic contraction on the norm of e_k , a term that scales with the noise in the gradient, and a term that grows with the noise in the Hessian:

$$\mathbb{E}[\|e_{k+1}\|^2 \mid e_k] = \|A e_k\|^2 + \eta^2 \mathbb{E}\|P \tilde{g}_k\|^2 + \eta^2 (e_k + d)' N (e_k + d). \quad (2)$$

Take the total expectation of (2). The gradient term is already constant, $\eta^2 \mathbb{E}\|P \tilde{g}_k\|^2 = \eta^2 \text{tr}(P \Sigma P)$; the other two are state-dependent, so bound each by its operator norm. The contraction gives $\mathbb{E}\|A e_k\|^2 \leq \|A\|_2^2 \mathbb{E}\|e_k\|^2$. For the Hessian term, expand $\mathbb{E}[(e_k + d)' N (e_k + d)] = \mathbb{E}[e_k' N e_k] + 2(\mathbb{E}e_k)' N d + d' N d$, where $\mathbb{E}[e_k' N e_k] \leq \|N\|_2 \mathbb{E}\|e_k\|^2$ and the cross term obeys $2(\mathbb{E}e_k)' N d \leq (\mathbb{E}e_k)' N (\mathbb{E}e_k) + d' N d \leq \|N\|_2 \mathbb{E}\|e_k\|^2 + d' N d$, using $\|\mathbb{E}e_k\|^2 \leq \mathbb{E}\|e_k\|^2$. Collecting the pieces gives the one-step affine bound

$$\mathbb{E}\|e_{k+1}\|^2 \leq \rho \mathbb{E}\|e_k\|^2 + \eta^2 (\text{tr}(P \Sigma P) + 2 d' N d). \quad (3)$$

Iterating (3) from e_0 gives the stated bound, and $\rho < 1$ sends its geometric term to 0, leaving the floor. $\square$

The rate ρ in Theorem 1 splits into a deterministic part $\|A\|_2^2$ and a multiplicative-noise part $2\eta^2\|N\|_2$, both functions of η ; how small can tuning η make it? Since $P\bar{H}$ is similar to the symmetric $P^{1/2}\bar{H}P^{1/2} \succ 0$, its eigenvalues $\lambda_i(P\bar{H})$ are real and positive, so A has eigenvalues $1 - \eta\lambda_i(P\bar{H})$. The largest of their magnitudes, $\max_i |1 - \eta\lambda_i(P\bar{H})|$, is minimized over η at $\eta = 2/(\lambda_{\min}(P\bar{H}) + \lambda_{\max}(P\bar{H}))$, where it equals $(\kappa - 1)/(\kappa + 1)$ with $\kappa = \kappa(P\bar{H})$, and is larger at every other η . Since $\|A\|_2$ is at least this magnitude,

$$\rho \geq \|A\|_2^2 \geq \left(\frac{\kappa - 1}{\kappa + 1}\right)^2 \quad \text{for every } \eta,$$

so the condition number $\kappa(P\bar{H})$ lower-bounds the rate at any step size, with equality only for a normal A at the optimal η and no Hessian noise.

3 Picking $P = \bar{H}^{-1}$ as a preconditioner

With $P = \bar{H}^{-1}$ the preconditioned Hessian is $P\bar{H} = I$, so $\kappa(P\bar{H}) = 1$, $A = (1 - \eta)I$, and the rate is $\rho = (1 - \eta)^2 + 2\eta^2\|N\|_2$. Minimizing over η gives $\eta = 1/(1 + 2\|N\|_2)$ and $\rho = 2\|N\|_2/(1 + 2\|N\|_2)$. The condition-number bound $((\kappa - 1)/(\kappa + 1))^2$ is now 0, so the rate is set entirely by the Hessian noise $N = \mathbb{E}[\tilde{H}_k \bar{H}^{-2} \tilde{H}_k]$. Theorem 1 specializes as follows.

Corollary 1. *Take $P = \bar{H}^{-1}$, and write $N = \mathbb{E}[\tilde{H}_k \bar{H}^{-2} \tilde{H}_k]$ and $\rho = (1 - \eta)^2 + 2\eta^2\|N\|_2$. If $\rho < 1$, then for every $k \geq 0$,*

$$\mathbb{E}\|e_k\|^2 \leq \rho^k \mathbb{E}\|e_0\|^2 + \frac{1 - \rho^k}{1 - \rho} \eta^2 (\text{tr}(\bar{H}^{-1} \Sigma \bar{H}^{-1}) + 2 d' N d).$$

Proof. Substitute $P = \bar{H}^{-1}$ into Theorem 1: $P\bar{H} = I$ gives $\|A\|_2^2 = (1 - \eta)^2$, $N = \mathbb{E}[\tilde{H}_k \bar{H}^{-2} \tilde{H}_k]$, and $\text{tr}(P\Sigma P) = \text{tr}(\bar{H}^{-1}\Sigma \bar{H}^{-1})$. $\square$

4 Preconditioning with the inverse of an independent Hessian sample

Now precondition with the inverse of an independent Hessian sample. Draw an invertible Q from the same distribution as the Hessian samples, independently of (g_k, H_k) and e_k and redrawn independently each step, and set $P = Q^{-1}$. The update uses this random P ,

$$x_{k+1} = x_k - \eta P \nabla f(x_k; g_k, H_k).$$

With P random, $A = I - \eta P \bar{H}$ is random and $N = \mathbb{E}[\tilde{H}_k P^2 \tilde{H}_k]$ now averages over P as well, so the rate of Theorem 1 reads

$$\rho \equiv \|\mathbb{E}[A'A]\|_2 + 2\eta^2 \|N\|_2.$$

Corollary 2. *If $\rho < 1$, then for every $k \geq 0$,*

$$\mathbb{E}\|e_k\|^2 \leq \rho^k \mathbb{E}\|e_0\|^2 + \frac{1 - \rho^k}{1 - \rho} \eta^2 (\mathbb{E}[\text{tr}(P\Sigma P)] + 2d'Nd),$$

so $\mathbb{E}\|e_k\|^2$ decays at rate ρ to a floor at most $\eta^2 (\mathbb{E}[\text{tr}(P\Sigma P)] + 2d'Nd)/(1 - \rho)$.

Proof. For a fixed realization of P the step is the deterministic- P update, so the conditional identity (2) holds. Averaging over P , independent of (g_k, H_k, e_k) , turns $\text{tr}(P\Sigma P)$ into $\mathbb{E}[\text{tr}(P\Sigma P)]$, keeps $N = \mathbb{E}[\tilde{H}_k P^2 \tilde{H}_k]$ (now averaging P too), and makes the contraction obey $\mathbb{E}\|Ae_k\|^2 \leq \|\mathbb{E}[A'A]\|_2 \mathbb{E}\|e_k\|^2$. The bounding and iteration of Theorem 1 then give the bound. $\square$

For comparison, plain SGD ($P = I$) has contraction $\|I - \eta \bar{H}\|_2^2$. To line the inverse-sample contraction up with it, split it into its mean and fluctuation,

$$\mathbb{E}[A'A] = (I - \eta \mathbb{E}[Q^{-1}]\bar{H})' (I - \eta \mathbb{E}[Q^{-1}]\bar{H}) + \eta^2 \bar{H} \text{Cov}(Q^{-1}) \bar{H}.$$

The first term is the $P = I$ contraction with $\bar{H}$ replaced by $\mathbb{E}[Q^{-1}]\bar{H}$; the second, absent under $P = I$, is the variance cost, growing with $\text{Cov}(Q^{-1})$ and diverging as Q nears singularity. Whether the replacement $\bar{H} \mapsto \mathbb{E}[Q^{-1}]\bar{H}$ helps is not something the model guarantees. Matrix Jensen gives only $\mathbb{E}[Q^{-1}] \succeq \bar{H}^{-1}$, hence $\mathbb{E}[Q^{-1}]\bar{H} \succeq I$; the gap $\mathbb{E}[Q^{-1}] - \bar{H}^{-1}$ vanishes only for deterministic Q and grows with the spread of Q . The replacement whitens, $\mathbb{E}[Q^{-1}]\bar{H} \rightarrow I$, only if Q concentrates near $\bar{H}$. When instead Q puts mass near singular matrices, $\mathbb{E}[Q^{-1}]$ runs far above $\bar{H}^{-1}$ and $\mathbb{E}[Q^{-1}]\bar{H}$ is *worse* conditioned than $\bar{H}$. So the inverse sample can lose on conditioning as well as on variance.

This variance falls on the contraction $\|\mathbb{E}[A'A]\|_2$ itself, not just the noise term, and the one-dimensional problem $\bar{H} = h$ shows it cleanly. A deterministic preconditioner makes the contraction $(1 - \eta h)^2$, which the step $\eta = 1/h$ sends to 0. The random inverse cannot: with $\mu_1 = \mathbb{E}[Q^{-1}]$ and $\mu_2 = \mathbb{E}[Q^{-2}]$, the contraction is a parabola in η ,

$$\mathbb{E}[A^2] = 1 - 2\eta h \mu_1 + \eta^2 h^2 \mu_2 \geq 1 - \frac{\mu_1^2}{\mu_2} > 0 \quad \text{for every } \eta,$$

its floor $1 - \mu_1^2/\mu_2$ reached at $\eta = \mu_1/(h\mu_2)$ and positive whenever Q is random (Cauchy–Schwarz, $\mu_1^2 < \mu_2$). So the variance of Q^{-1} keeps the contraction, and hence the rate, bounded away from 0 at any step size, before any gradient or Hessian noise is added. That is what a deterministic preconditioner avoids and a random one cannot.

5 Preconditioning with the gradient's own Hessian

Take instead $P = H_k^{-1}$, the very Hessian in the gradient (assume H_k invertible), and substitute it into the error recurrence (1). Its multiplicative Hessian term recombines with $A = I - \eta H_k^{-1} \bar{H}$: the e_k -coefficient collapses, $I - \eta H_k^{-1} \bar{H} - \eta H_k^{-1} \tilde{H}_k = I - \eta H_k^{-1} H_k = (1 - \eta)I$, while the additive terms combine into $-\eta H_k^{-1}(\tilde{g}_k + \tilde{H}_k d) = -\eta H_k^{-1} \nabla f(x^*; g_k, H_k)$, the stochastic gradient at the optimum. So (1) becomes

$$e_{k+1} = (1 - \eta)e_k - \eta H_k^{-1} \nabla f(x^*; g_k, H_k).$$

After the cancellation this is (1) of the first section, a contraction plus additive noise with no multiplicative Hessian term, except that the noise $H_k^{-1} \nabla f(x^*)$ is not centered. Splitting it into mean and fluctuation, $\mu \equiv \mathbb{E}[H_k^{-1} \nabla f(x^*)]$ and $\zeta_k \equiv H_k^{-1} \nabla f(x^*) - \mu$ (ζ_k zero mean, i.i.d., independent of e_k), the mean appears as an offset:

$$e_{k+1} = (1 - \eta)e_k - \eta\mu - \eta\zeta_k. \quad (4)$$

The offset μ is a bias: the gradient at x^* is unbiased, $\mathbb{E}[\nabla f(x^*)] = 0$, yet preconditioning by the correlated H_k^{-1} makes $\mu \neq 0$. With $g_k \perp H_k$, since $\nabla f(x^*) = g_k - H_k \bar{H}^{-1} \bar{g}$, $\mu = (\mathbb{E}[H_k^{-1}] - \bar{H}^{-1})\bar{g}$, nonzero by Jensen $\mathbb{E}[H_k^{-1}] \succeq \bar{H}^{-1}$.

Removing the offset recovers the first section's recurrence exactly: the deviation $e_k + \mu$ obeys $(e_{k+1} + \mu) = (1 - \eta)(e_k + \mu) - \eta\zeta_k$, which is (1) with $A = (1 - \eta)I$, $N = 0$, and zero-mean noise ζ_k . So Theorem 1 specializes as follows.

Corollary 3. *With $\rho = (1 - \eta)^2$, for every $k \geq 0$,*

$$\mathbb{E}\|e_k + \mu\|^2 \leq \rho^k \mathbb{E}\|e_0 + \mu\|^2 + \frac{1 - \rho^k}{1 - \rho} \eta^2 \text{tr Cov}(\zeta_k),$$

and $\mathbb{E}[e_k] \rightarrow -\mu$, so $\mathbb{E}\|e_k\|^2 \rightarrow \|\mu\|^2 + \frac{\eta}{2 - \eta} \text{tr Cov}(\zeta_k)$.

Proof. The deviation $e_k + \mu$ satisfies Theorem 1's recurrence with $A = (1 - \eta)I$, $N = 0$, and gradient covariance $\text{Cov}(\zeta_k)$, giving the bound and $\mathbb{E}[e_k + \mu] \rightarrow 0$; then $\mathbb{E}\|e_k\|^2 = \mathbb{E}\|e_k + \mu\|^2 + \text{tr Cov}(e_k)$ gives the limit. $\square$

So the iterate converges to $x^* - \mu$, not x^* : the bias $\|\mu\|^2$ is present for every η and survives $\eta \rightarrow 0$, unremovable by any step size. The independent sample of the previous section avoids it, since there P shares nothing with the gradient and $\mathbb{E}[P \nabla f(x^*)] = \mathbb{E}[P] \mathbb{E}[\nabla f(x^*)] = 0$.

Remark 1 (The floor in $\bar{g}$). *With $g_k \perp H_k$, both parts of the stationary error are exact quadratics in the mean gradient. The bias is $\|\mu\|^2 = \bar{g}'(\mathbb{E}[H_k^{-1}] - \bar{H}^{-1})^2 \bar{g}$, and the fluctuation is*

$$\text{tr Cov}(\zeta_k) = \bar{g}' V \bar{g} + c,$$

with $V = \mathbb{E}[(H_k^{-1} - \mathbb{E}[H_k^{-1}])^2]$ the covariance of the inverse Hessian and $c = \text{tr}(\mathbb{E}[H_k^{-2}] \Sigma)$ the variance of the preconditioned gradient noise $H_k^{-1} \tilde{g}_k$.